

HUMBER ENGINEERING

MENG-3020

SYSTEMS MODELING & SIMULATION

LECTURE 5

LECTURE 5

Electrical Systems

- Modeling of Electrical Systems
 - Variables, Elements & Element Laws
 - Interconnection Laws
 - Complex Impedance Method
 - Operational Amplifiers

Electrical Systems: Variables & Elements

- The **variables** that are used to describe the electrical systems are:

- $v(t)$: **Voltage (V)**
- $i(t)$: **Current (A)**

- All these variables are function of time.

- **Current** is defined as the rate of change of the charge $Q(t)$ passing through an area:

$$i(t) = \frac{dQ(t)}{dt}$$

- **Voltage** is defined as the required work or energy to move a charge between two points in a circuit.

- Electric circuit **elements** may be classified as passive and active elements.

- The passive elements can store or dissipate energy that is already present in the circuit, but they cannot introduce additional energy into the circuit:

- **Resistance Elements: Resistor**
- **Capacitance Elements: Capacitor**
- **Inductance Elements: Inductor**

- The active elements can introduce energy into the circuit:

- **Voltage Sources**
- **Current Sources**
- **Op-Amps**

Electrical Systems Element Laws

□ Resistance Elements: Resistor

- A **resistor** is an element for which there is an algebraic relationship between the voltage across its terminals and the current through it.
- In a linear resistor the voltage and current are directly proportional to each other by **Ohm's law**.
- The R is the resistance. The unit is ohm (Ω).

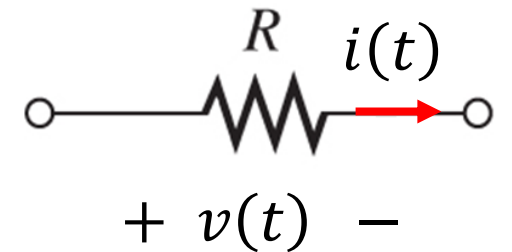
$$v(t) = Ri(t)$$

- The resistance of a body of length l and constant cross-sectional area A made of a material with resistivity ρ is:

$$R = \rho \frac{l}{A}$$

- Resistors do not store electric energy in any form but instead dissipate it as heat.
- We can write the power dissipated by a linear resistor as:

$$P = Ri^2 \quad \text{or} \quad P = \frac{v^2}{R}$$

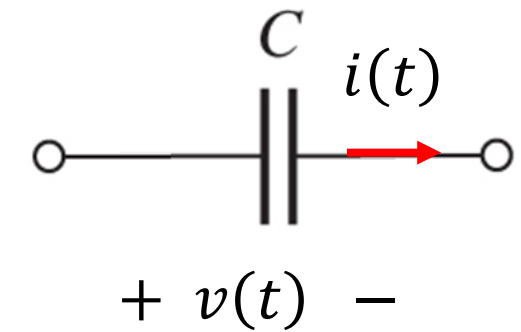


Electrical Systems Element Laws

□ Capacitance Elements: Capacitor

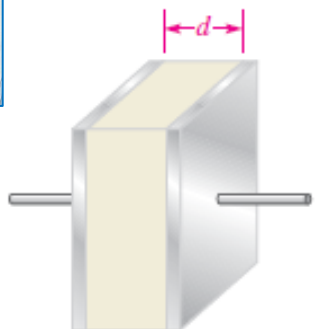
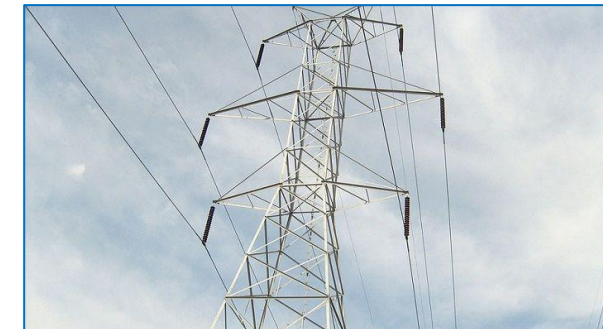
- A **capacitor** is an element that obeys an algebraic relationship between the voltage and the charge.
- For a linear capacitor, the charge and voltage are related as below
- The C is the capacitance. The unit is farad (F).

$$Q(t) = Cv(t) \rightarrow i(t) = C \frac{dv(t)}{dt} \rightarrow v(t) = \frac{1}{C} \int i(t) dt$$



- The capacitance of a capacitor depends on the plate area A , the distance between the plates d , and the permittivity of dielectric ϵ :

$$C = \frac{\epsilon A}{d}$$



- The **energy** supplied to a capacitor is stored in its electrical field.
- For a fixed linear capacitor, the **stored energy** is:

$$W = \frac{1}{2} C v^2$$

Electrical Systems Element Laws

□ Inductance Elements: Inductor

- An **inductor** is an element for which there is an algebraic relationship between the voltage across its terminals and the derivative of the flux linkage.
- For a **linear inductor**, the current and voltage are related as below
- The L is the **inductance**. The unit is **henry (H)**.

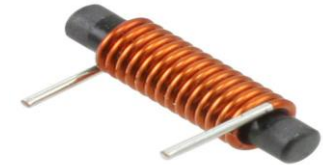
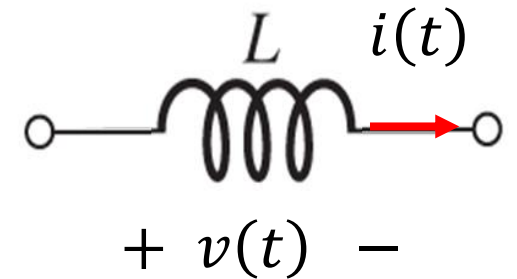
$$v(t) \propto \frac{d\phi}{dt} \rightarrow v(t) = L \frac{di(t)}{dt} \rightarrow i(t) = \frac{1}{L} \int v(t) dt$$

- The inductance of an inductor depends on the **number of turns of wire N** , the **length of the core l** , the **cross-sectional area of the core A** , and the **permeability of the core material μ** :

$$L = N^2 \frac{\mu A}{l}$$

- The **energy** supplied to an **inductor** is stored in its **magnetic field**.
- For a fixed linear inductor, the **stored energy** is:

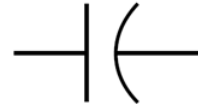
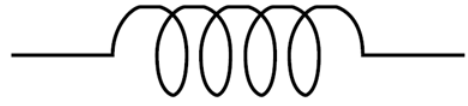
$$W = \frac{1}{2} Li^2$$



Electrical Systems Element Laws

□ Summary

- Table shows summary of the **voltage-current** and **current-voltage** relationships for capacitor, resistor, and inductor.

Component	Voltage-current	Current-voltage
 Capacitor	$v(t) = \frac{1}{C} \int_0^1 i(\tau) d\tau$	$i(t) = C \frac{dv(t)}{dt}$
 Resistor	$v(t) = Ri(t)$	$i(t) = \frac{1}{R} v(t)$
 Inductor	$v(t) = L \frac{di(t)}{dt}$	$i(t) = \frac{1}{L} \int_0^1 v(\tau) d\tau$

Electrical Systems: Interconnection Laws

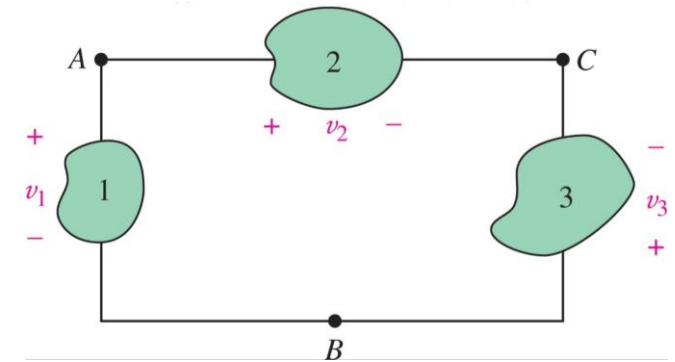
- Two interconnection laws are used in modeling electrical circuits.

□ Kirchhoff's Voltage Law (KVL)

- The algebraic sum of the voltages around any loop in an electrical circuit is zero.

$$\sum v_j = 0 \quad \text{around any loop}$$

where v_j denotes the voltage across the j th element in the loop.



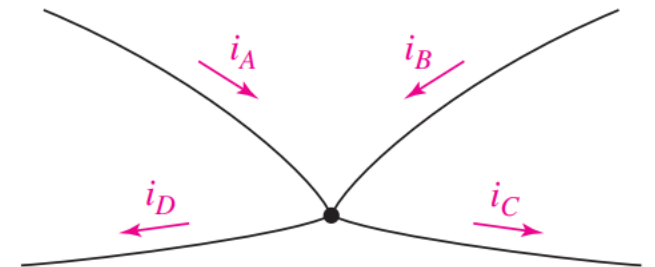
$$v_2 - v_3 - v_1 = 0$$

□ Kirchhoff's Current Law (KCL)

- The algebraic sum of all currents entering and leaving a node is zero.

$$\sum i_j = 0 \quad \text{at any node}$$

where the summation is over the currents through all the elements joined to the node.



$$i_A + i_B + (-i_C) + (-i_D) = 0$$

□ Impedance Method

- Replace the component values with their impedance values in Laplace domain.
- Apply the circuit laws KVL and KCL in Laplace domain to find the transfer function model.

Modeling of Electrical Systems

Example 1

Given RLC network,

a) Find the differential equation and the transfer function model relating the capacitor voltage $v_c(t)$ to the input voltage $v(t)$.

First, select the input and output of the system.

The applied voltage $v(t)$ is the input. The capacitor voltage $v_c(t)$ is the output.

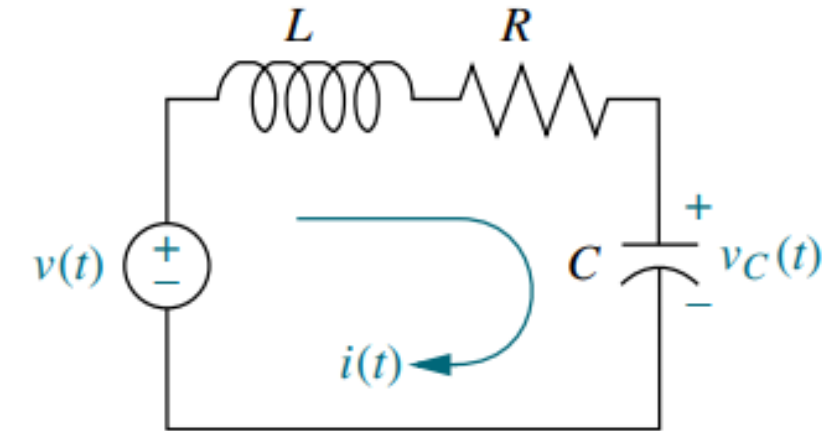
By using the **Kirchhoff's Voltage Law (KVL)** we have:

$$v(t) = v_L(t) + v_R(t) + v_C(t)$$

The **differential equation** relating $v(t)$ to $v_c(t)$ is determined as

$$v(t) = L \frac{di(t)}{dt} + Ri(t) + v_c(t) \quad \xrightarrow{i(t)=C \frac{dv_c(t)}{dt}} \quad v(t) = LC \frac{d^2 v_c(t)}{dt^2} + RC \frac{dv_c(t)}{dt} + v_c(t)$$

**Second-order
Differential Equation**



Taking **Laplace transform** of both side by assuming the **zero initial conditions**

$$V(s) = LCs^2 V_c(s) + RCs V_c(s) + V_c(s) \rightarrow V(s) = (LCs^2 + RCs + 1)V_c(s)$$

$$\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}$$

**Transfer Function
Model**

Modeling of Electrical Systems

Example 1

Given RLC network,

b) Find the differential equation and the transfer function model relating the current $i(t)$ to the input voltage $v(t)$.

In this part, current $i(t)$ is the output. The applied voltage $v(t)$ is the input.

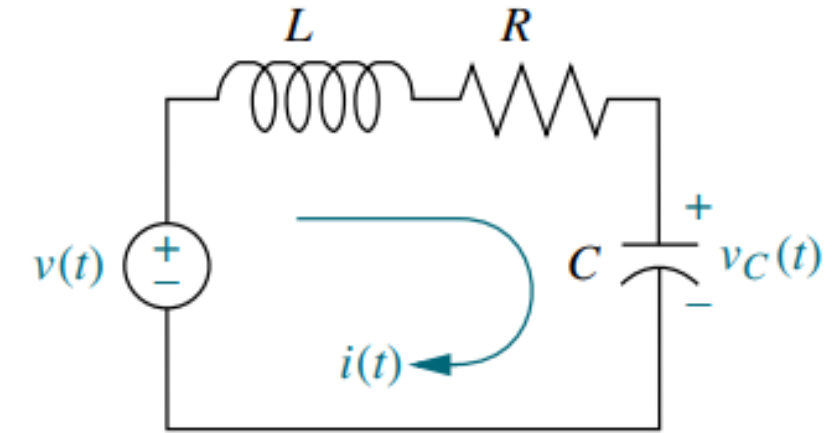
By using the **Kirchhoff's Voltage Law (KVL)** we have:

$$v(t) = v_L(t) + v_R(t) + v_C(t)$$

The **differential equation** relating $v(t)$ to $i(t)$ is determined as

$$v(t) = L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int i(t) dt \rightarrow \frac{dv(t)}{dt} = L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t)$$

**Second-order
Differential Equation**



Taking **Laplace transform** of both side by assuming the **zero initial conditions**

$$sV(s) = Ls^2 I(s) + RsI(s) + \frac{1}{C} I(s) \rightarrow CsV(s) = (LCs^2 + RCs + 1)I(s)$$

$$\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}$$

**Transfer Function
Model**

Modeling of Electrical Systems

Example 1

Given RLC network,

c) Having the transfer function model and given component values find the loop current $i(t)$ if the applied voltage is $v(t) = 10V$, $t \geq 0$.

$$L = 1H, \quad R = 3\Omega, \quad C = 0.5F$$

Recall the **current-voltage transfer function** from Step (b):

$$\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}$$

First, find the transfer function from the given values $\rightarrow \frac{I(s)}{V(s)} = \frac{0.5s}{0.5s^2 + 1.5s + 1}$

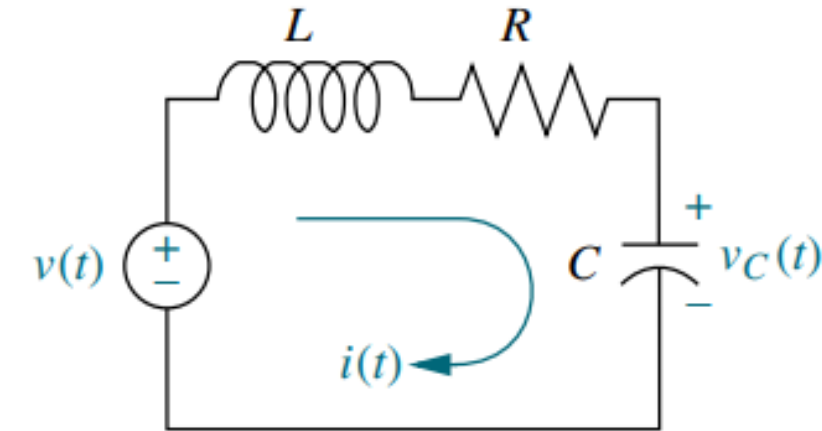
Next, find the loop **current** $I(s)$.

$$I(s) = \left(\frac{0.5s}{0.5s^2 + 1.5s + 1} \right) V(s) = \left(\frac{0.5s}{0.5s^2 + 1.5s + 1} \right) \left(\frac{10}{s} \right) = \frac{5}{0.5s^2 + 1.5s + 1} = \frac{10}{s^2 + 3s + 2} = \frac{10}{(s+1)(s+2)}$$

Apply **partial fraction expansion** method $\rightarrow I(s) = \frac{10}{s+1} + \frac{-10}{s+2}$

Take the inverse Laplace transform to find $i(t)$ assuming **zero initial conditions**.

$$i(t) = 10e^{-t} - 10e^{-2t} = 10(e^{-t} - e^{-2t}), \quad t \geq 0$$



Modeling of Electrical Systems

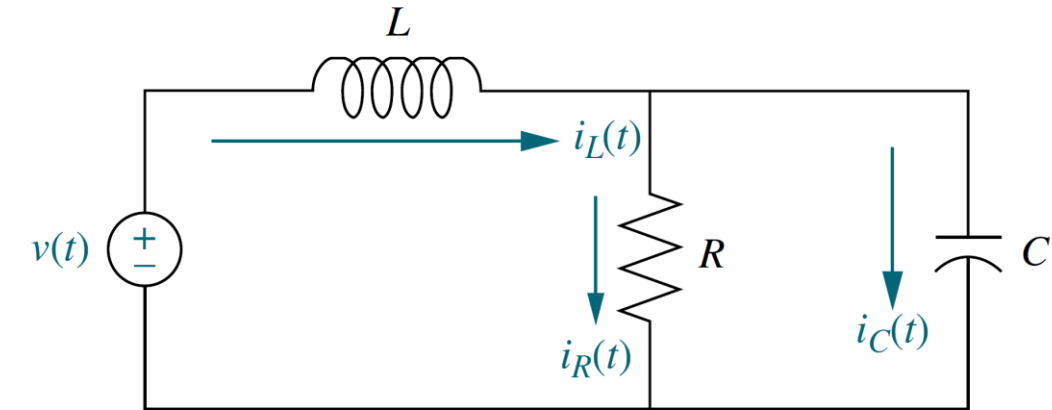
Example 2

Given the electric network, find a state-space model if the input is the applied voltage $v(t)$ and output is the current through the resistor $i_R(t)$. Draw the block-diagram model.

The **state variables** q_1 and q_2 are selected as the **inductor current** $i_L(t)$ and the **capacitor voltage** $v_c(t)$.

$$q_1 = i_L \rightarrow \dot{q}_1 = \dot{i}_L$$

$$q_2 = v_c \rightarrow \dot{q}_2 = \dot{v}_c$$



By using a **KCL** and a **KVL** we have:

$$i_L = i_R + i_C \rightarrow i_L = \frac{v_R}{R} + C\dot{v}_c \rightarrow i_L = \frac{1}{R}v_c + C\dot{v}_c \rightarrow \dot{v}_c = \frac{1}{C}i_L - \frac{1}{RC}v_c \rightarrow \dot{q}_2 = \frac{1}{C}q_1 - \frac{1}{RC}q_2$$

$$v(t) = v_L + v_c \rightarrow v(t) = Li_L + v_c \rightarrow i_L = \frac{1}{L}v(t) - \frac{1}{L}v_c \rightarrow \dot{q}_1 = \frac{1}{L}v(t) - \frac{1}{L}q_2$$

State-variable equations are obtained as:

$$\begin{cases} \dot{q}_1 = \frac{1}{L}v(t) - \frac{1}{L}q_2 \\ \dot{q}_2 = \frac{1}{C}q_1 - \frac{1}{RC}q_2 \end{cases}$$

The **output equation** is obtained as:

$$y(t) = i_R = \frac{v_R}{R} = \frac{v_c}{R} \rightarrow y(t) = \frac{1}{R}q_2$$

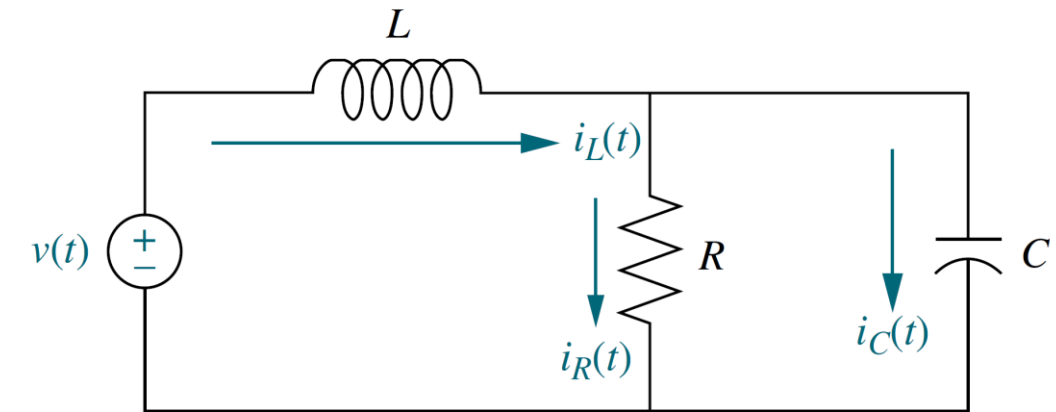
Modeling of Electrical Systems

Example 2

Given the electric network, find a state-space model if the input is the applied voltage $v(t)$ and output is the current through the resistor $i_R(t)$. Draw the block-diagram model.

Having the state and output equations:

$$\begin{cases} \dot{q}_1 = \frac{1}{L}v(t) - \frac{1}{L}q_2 \\ \dot{q}_2 = \frac{1}{C}q_1 - \frac{1}{RC}q_2 \\ y(t) = \frac{1}{R}q_2 \end{cases}$$



We can represent the **state equations** and the **output equation** in the standard matrix-vector form as below:

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$

State Equation



$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

Output Equation

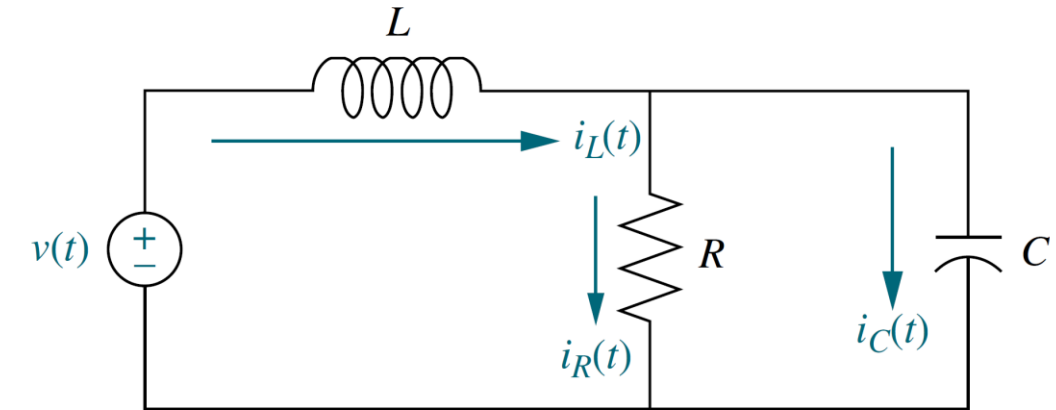


$$y(t) = \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + [0]v(t)$$

Modeling of Electrical Systems

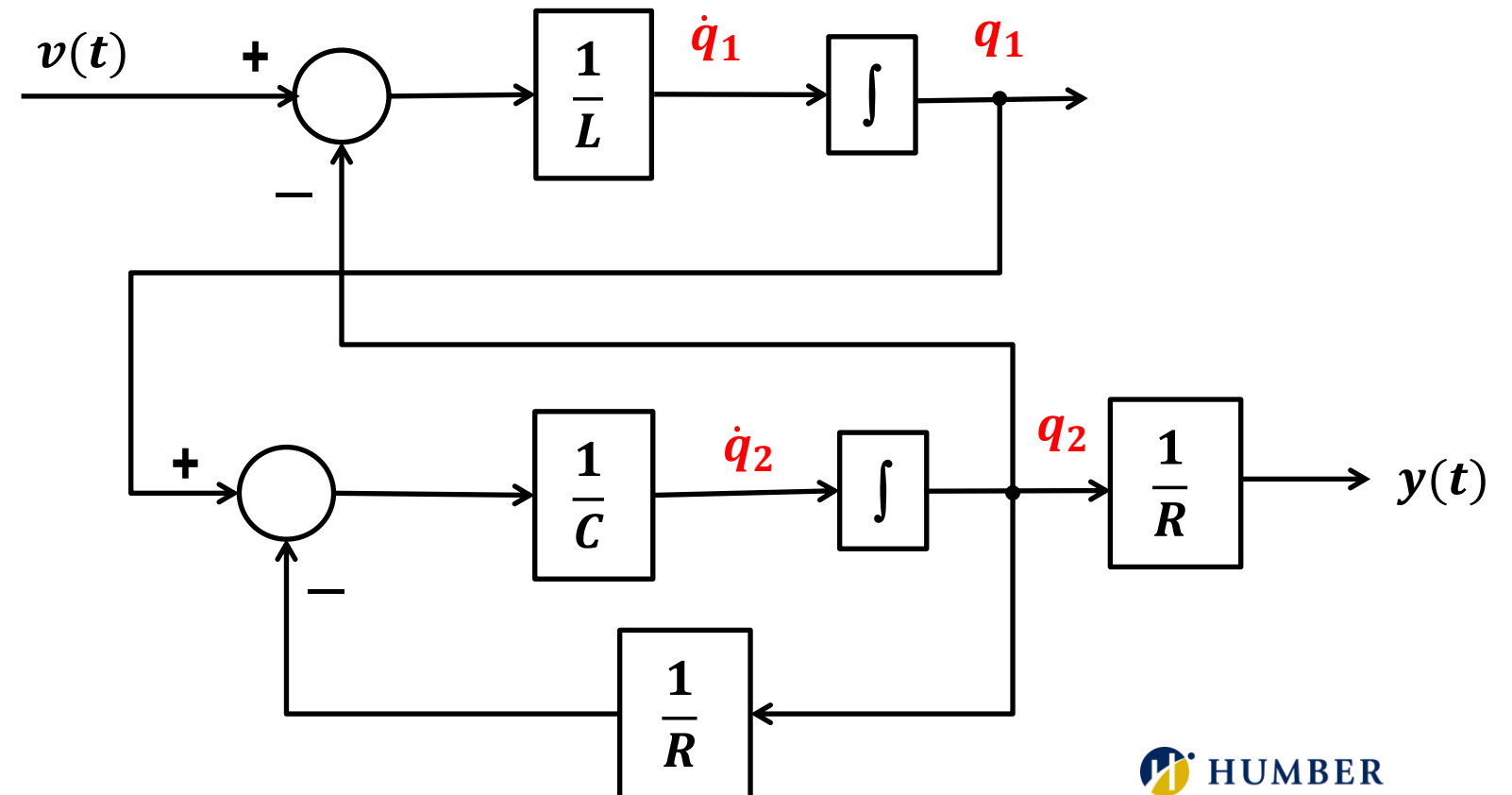
Example 2

Given the electric network, find a state-space model if the input is the applied voltage $v(t)$ and output is the current through the resistor $i_R(t)$. Draw the block-diagram model.



Following block diagram visualizes the state variables and the system outputs.

$$\begin{cases} \dot{q}_1 = \frac{1}{L}v(t) - \frac{1}{L}q_2 & \rightarrow & \dot{q}_1 = \frac{1}{L}(v(t) - q_2) \\ \dot{q}_2 = \frac{1}{C}q_1 - \frac{1}{RC}q_2 & \rightarrow & \dot{q}_2 = \frac{1}{C}\left(q_1 - \frac{1}{R}q_2\right) \\ y(t) = \frac{1}{R}q_2 \end{cases}$$



Modeling of Electrical Systems

□ Impedance Method

- In deriving **transfer functions** for electrical circuits, we can write the Laplace-transformed equations **directly**, without writing the differential equations, by using the **complex impedance** in **Laplace domain**.

- Recall the voltage-current relation of the electrical components, we can define the **complex impedance $Z(s)$ in Laplace domain** for each element as shown below:

- Impedance of a **capacitor**:

$$i(t) = C \frac{dv(t)}{dt} \rightarrow I(s) = CsV(s) \rightarrow Z(s) = \frac{V(s)}{I(s)} = \frac{1}{Cs}$$

- Impedance of an **inductor**:

$$v(t) = L \frac{di(t)}{dt} \rightarrow V(s) = LsI(s) \rightarrow Z(s) = \frac{V(s)}{I(s)} = Ls$$

- Impedance of a **resistor**:

$$v(t) = Ri(t) \rightarrow V(s) = RI(s) \rightarrow Z(s) = \frac{V(s)}{I(s)} = R$$

$$Z(s) = \frac{V(s)}{I(s)}$$

Impedance (ohms) Voltage (V) Current (A)

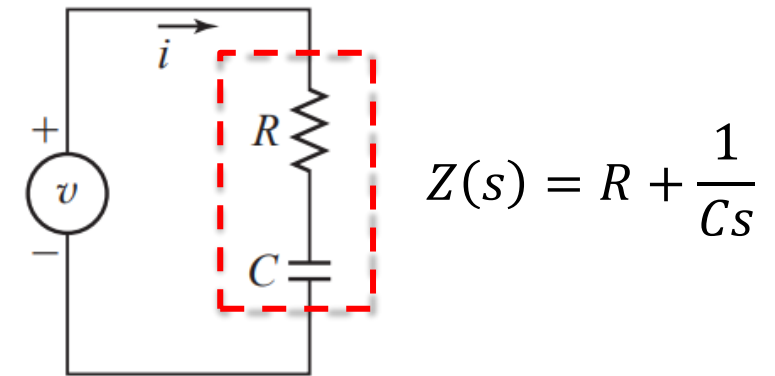
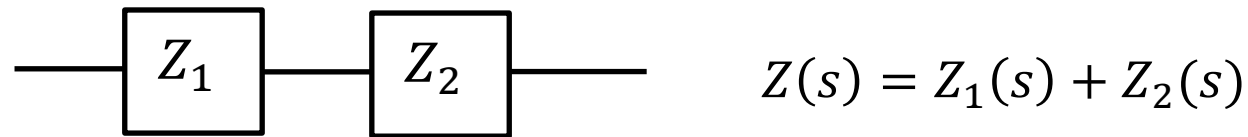
Modeling of Electrical Systems

□ Series & Parallel Impedances

- The concept of impedance is useful because the impedances of individual elements can be combined with **series** and **parallel** laws to find the impedance at any point in the system.

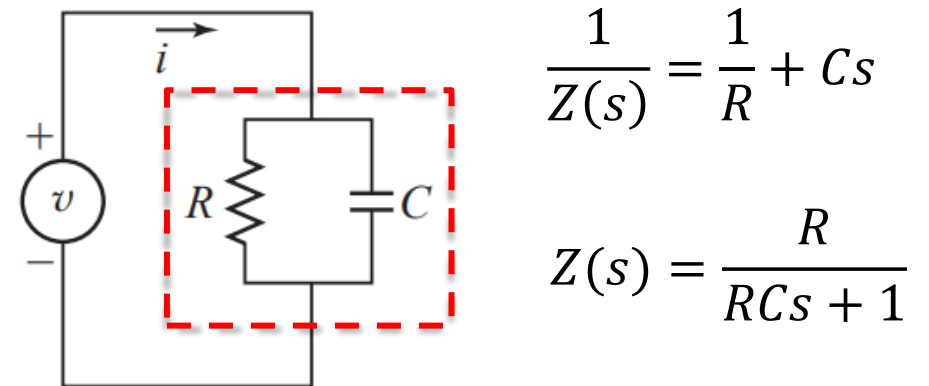
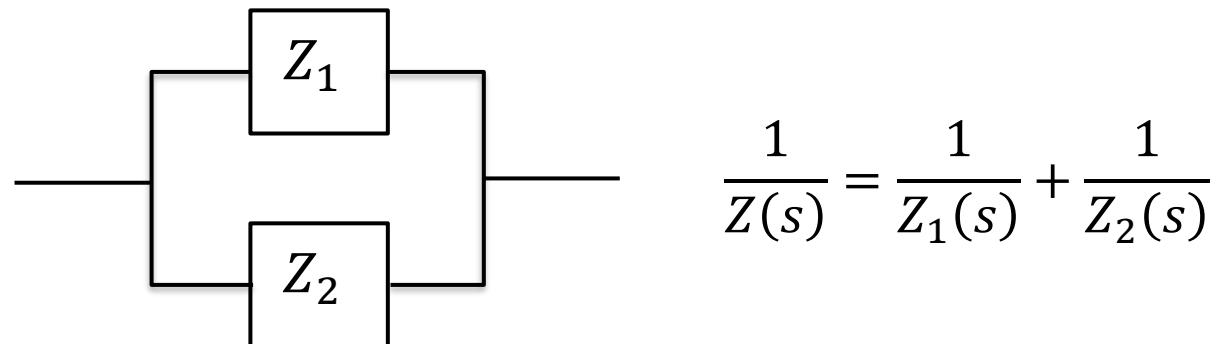
■ Series Impedances

- Two impedances are in **series** if they have the same current.
- The total impedance is the **sum** of the individual impedances.



■ Parallel Impedances

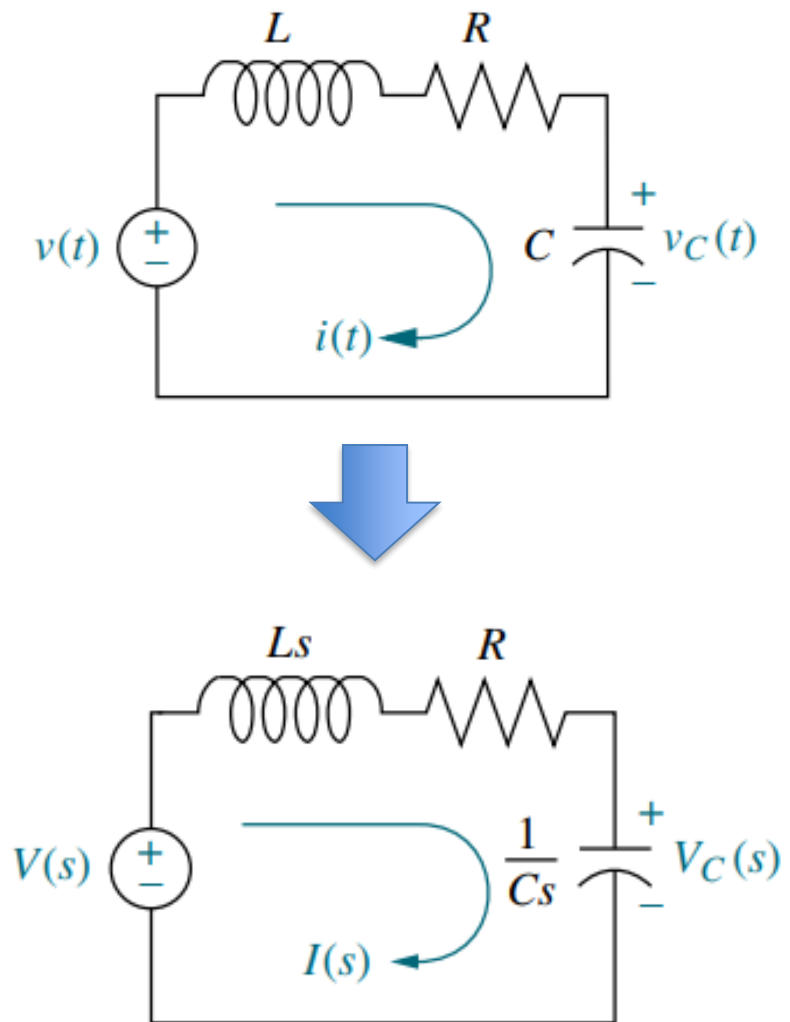
- The impedances are in **parallel**, if have the same voltage difference across them.
- Their impedances combine by the **reciprocal** rule.



Modeling of Electrical Systems

□ Circuit Analysis Using Impedance Method

- To find the **transfer function** model of electrical networks using complex **impedance method**, we can perform the following steps:
 - Replace the **passive element** values with their **impedances**.
 - Replace all **sources** and **time variables** with their **Laplace transform**.
 - Apply circuit laws such as, **KVL**, **KCL**, **voltage division**, **current division**, **mesh analysis** or **nodal analysis**, ...
 - Solve the simultaneous equations for the **output**.
 - Form the **transfer function**.



Modeling of Electrical Systems

□ Complex Impedance Method

- The following example shows how the concept of impedance simplifies the solution for the transfer functions:

$$G_1(s) = \frac{I(s)}{V(s)} \qquad G_2(s) = \frac{V_c(s)}{V(s)}$$

- Apply **KVL** in the loop in Laplace domain:

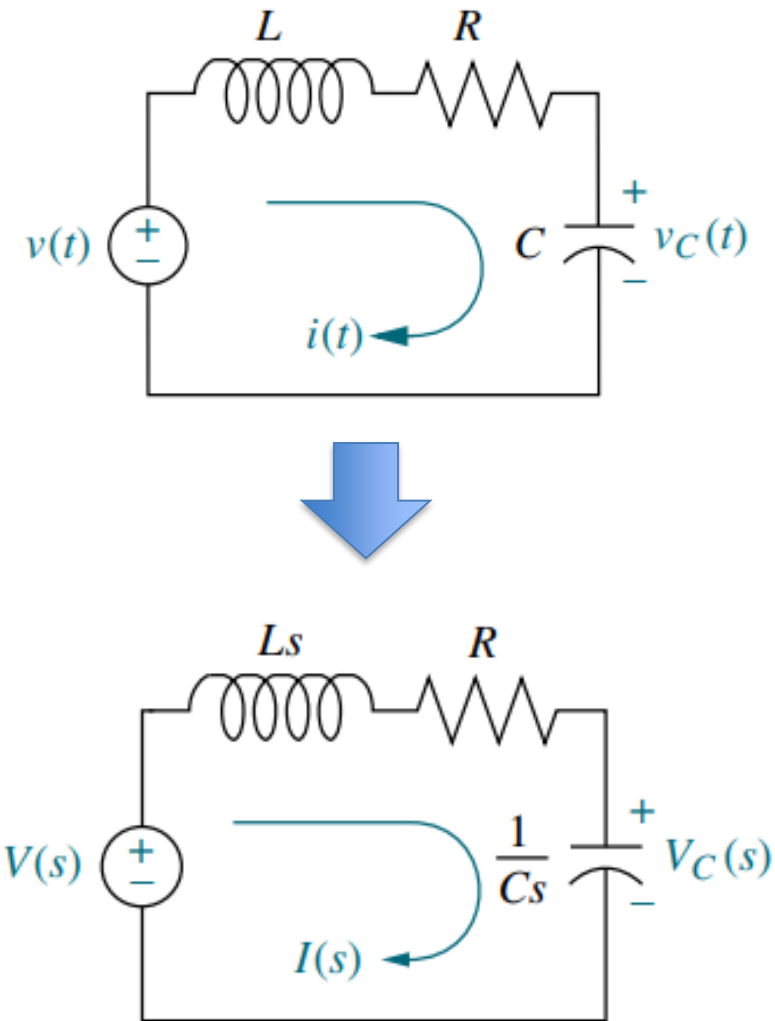
KVL $\rightarrow V(s) = V_L(s) + V_R(s) + V_c(s)$

$$V(s) = LsI(s) + RI(s) + \frac{1}{Cs}I(s) \quad \rightarrow \quad V(s) = \left(Ls + R + \frac{1}{Cs} \right) I(s)$$

$$\rightarrow V(s) = \left(\frac{LCs^2 + RCs + 1}{Cs} \right) I(s) \quad \rightarrow \quad \boxed{\frac{I(s)}{V(s)} = \frac{Cs}{LCs^2 + RCs + 1}}$$

- Apply **voltage-division** in Laplace domain:

$$V_c(s) = \left(\frac{\frac{1}{Cs}}{Ls + R + \frac{1}{Cs}} \right) V(s) = \left(\frac{\frac{1}{Cs}}{\frac{LCs^2 + RCs + 1}{Cs}} \right) V(s) \quad \rightarrow \quad \boxed{\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}}$$



Modeling of Electrical Systems

Example 3 Given the electric network,
a) Find the transfer function $I_2(s)/V(s)$ applying the impedance method.

First step is to convert the circuit into Laplace domain for impedances and variables, assuming zero initial conditions.

Then apply the mesh analysis for each loop:

Mesh 1: $R_1 I_1(s) + Ls(I_1(s) - I_2(s)) - V(s) = 0$

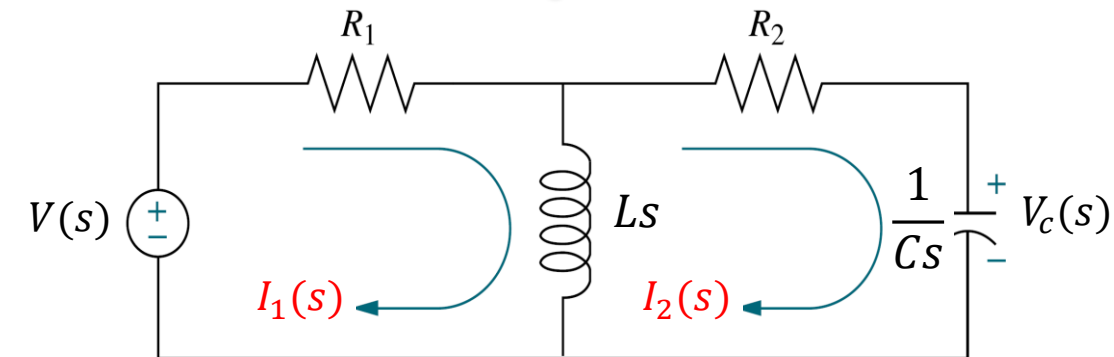
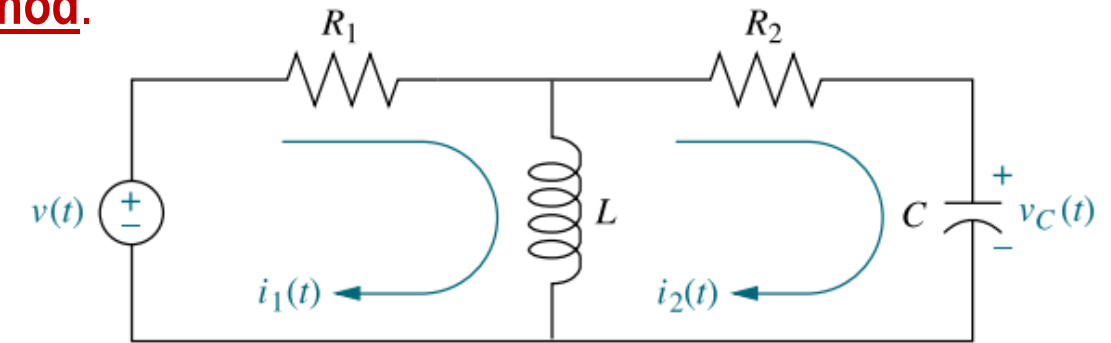
$$(R_1 + Ls)I_1(s) - LsI_2(s) = V(s) \quad \text{Eqn. (1)}$$

Mesh 2: $R_2 I_2(s) + \frac{1}{Cs}I_2(s) + Ls(I_2(s) - I_1(s)) = 0$

$$\left(R_2 + \frac{1}{Cs} + Ls\right)I_2(s) - LsI_1(s) = 0 \quad \text{Eqn. (2)}$$

Solve Eqn. (1) and (2) for transfer function $I_2(s)/V(s)$:

$$\begin{cases} (R_1 + Ls)I_1(s) - LsI_2(s) = V(s) \\ \left(R_2 + \frac{1}{Cs} + Ls\right)I_2(s) - LsI_1(s) = 0 \end{cases}$$



Modeling of Electrical Systems

Example 3

Given the electric network,

a) Find the transfer function $I_2(s)/V(s)$ applying the impedance method.

$$\begin{cases} (R_1 + Ls)I_1(s) - LsI_2(s) = V(s) \\ \left(R_2 + \frac{1}{Cs} + Ls\right)I_2(s) - LsI_1(s) = 0 \end{cases}$$

From Eqn. (2) we have $\rightarrow I_1(s) = \frac{1}{Ls} \left(R_2 + \frac{1}{Cs} + Ls\right) I_2(s) = \left(\frac{R_2}{Ls} + \frac{1}{LCs^2} + 1\right) I_2(s)$

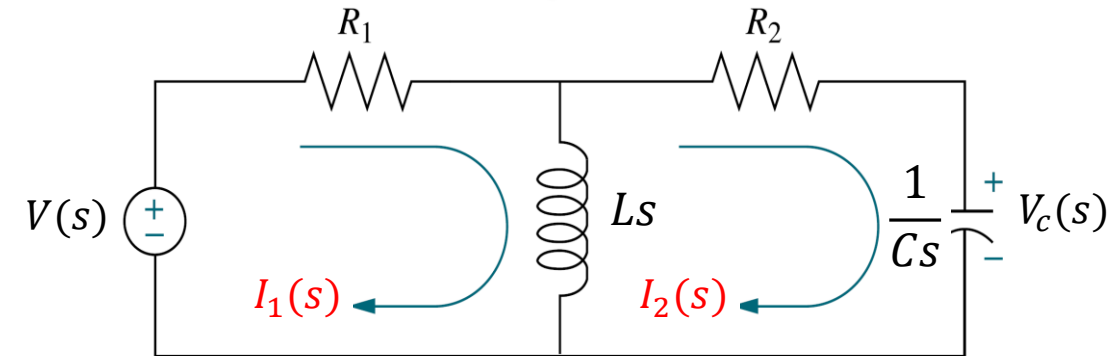
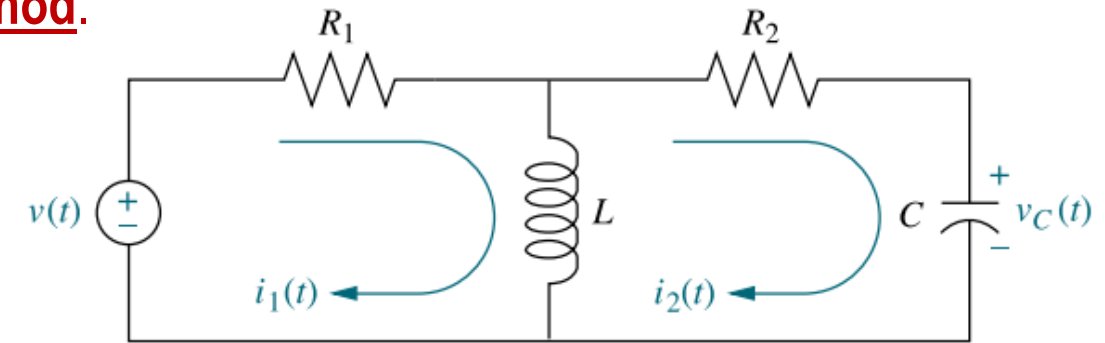
Substitute in Eqn. (1) $\rightarrow (R_1 + Ls) \left(\frac{R_2}{Ls} + \frac{1}{LCs^2} + 1\right) I_2(s) - LsI_2(s) = V(s)$

$$\left(\frac{R_1R_2}{Ls} + \frac{R_1}{LCs^2} + R_1 + R_2 + \frac{1}{Cs} + Ls - Ls\right) I_2(s) = V(s)$$

$$\left(\frac{R_1R_2Cs + R_1 + (R_1 + R_2)LCs^2 + Ls}{LCs^2}\right) I_2(s) = V(s)$$

Form the transfer function $I_2(s)/V(s) \rightarrow$

$$\frac{I_2(s)}{V(s)} = \frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$



Transfer Function Model

Modeling of Electrical Systems

Example 3

Given the electric network,

b) Having the transfer function $I_2(s)/V(s)$, find the transfer function $V_c(s)/V(s)$.

Recall the transfer function $I_2(s)/V(s)$ from Part (a):

$$\frac{I_2(s)}{V(s)} = \frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$

Having relationship between $i_2(t)$ and $v_c(t)$:

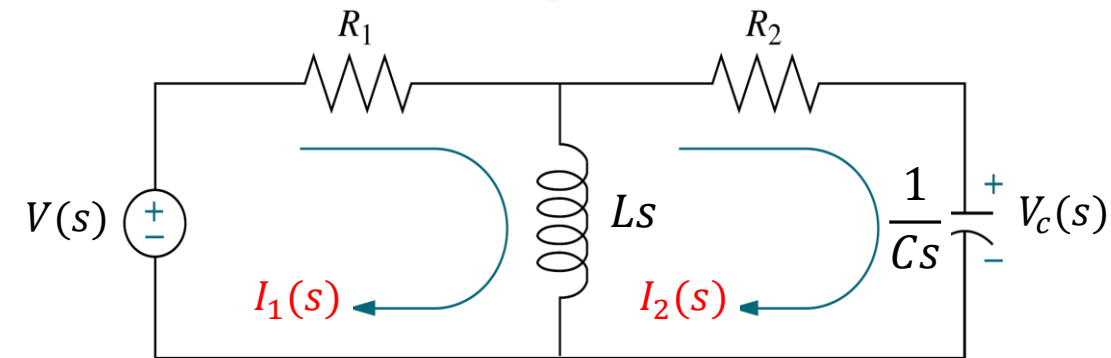
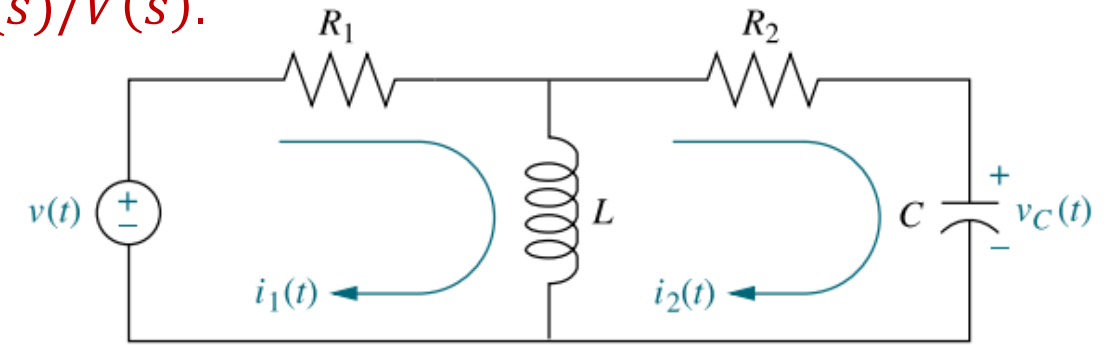
$$i_2(t) = C \frac{dv_c(t)}{dt} \rightarrow I_2(s) = CsV_c(s)$$

Replace $I_2(s)$ in the transfer function of $I_2(s)/V(s)$ and simplify it to obtain the transfer function of $V_c(s)/V(s)$:

$$\frac{CsV_c(s)}{V(s)} = \frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$

$$\frac{V_c(s)}{V(s)} = \frac{Ls}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$

Transfer Function Model



Modeling of Electrical Systems

Example 3 Given the electric network,
c) Having the transfer function $V_c(s)/V(s)$ find the differential equation model of the system

Recall the transfer function $V_c(s)/V(s)$ from Part (b):

$$\frac{V_c(s)}{V(s)} = \frac{Ls}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$

Rearrange the transfer function:

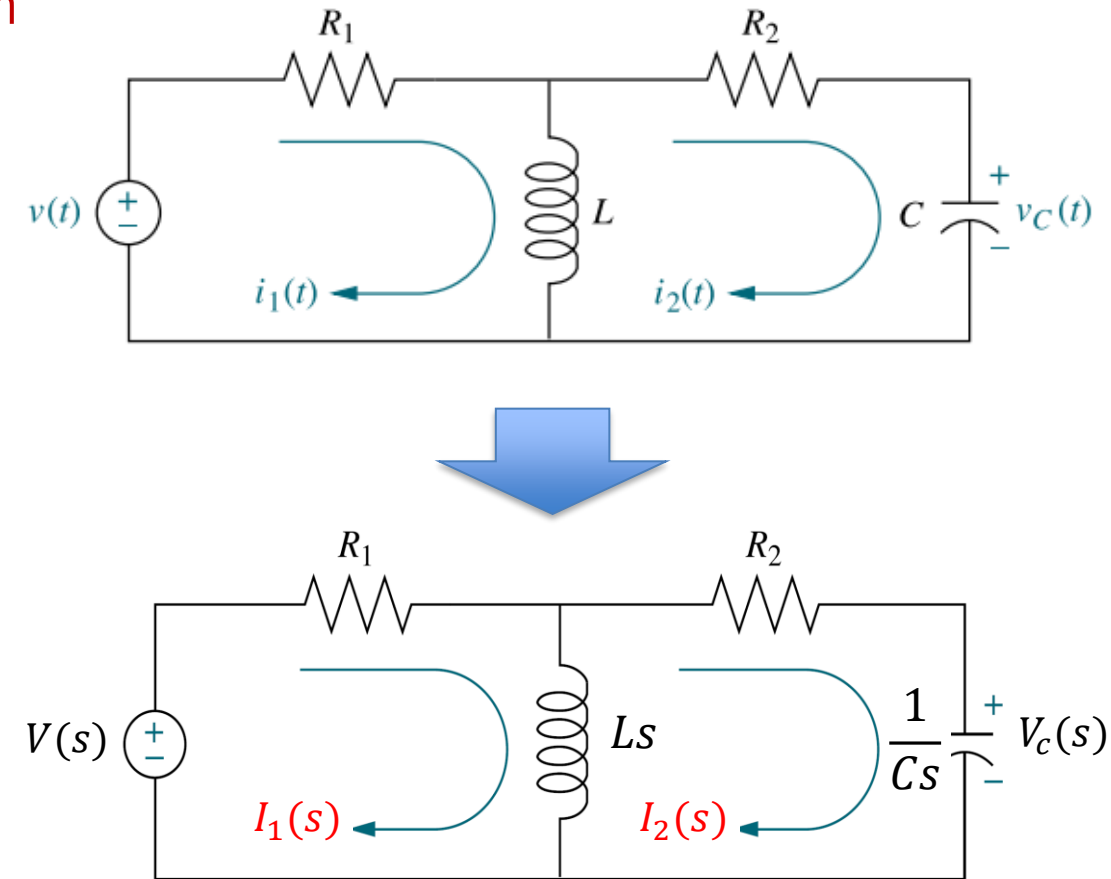
$$((R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1)V_c(s) = LsV(s)$$

$$(R_1 + R_2)LCs^2V_c(s) + (R_1R_2C + L)sV_c(s) + R_1V_c(s) = LsV(s)$$

Take inverse Laplace transform assuming zero initial conditions.

$$(R_1 + R_2)LC\ddot{v}_c(t) + (R_1R_2C + L)\dot{v}_c(t) + R_1v_c(t) = L\dot{v}(t)$$

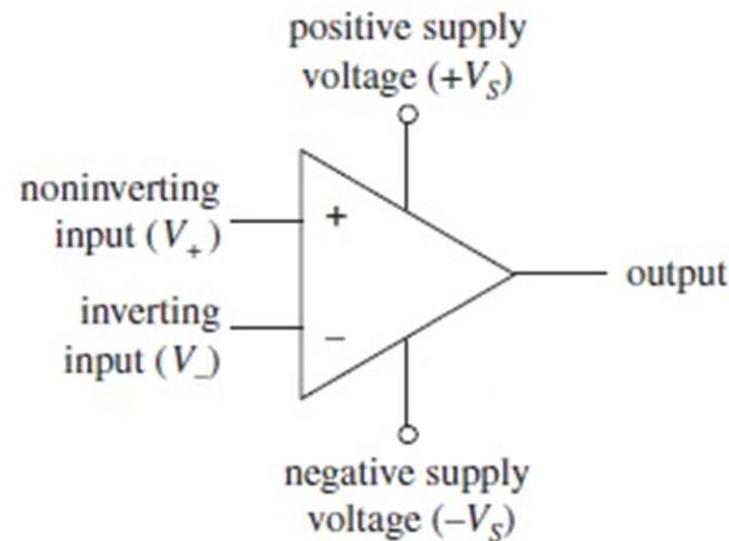
Differential Equation Model



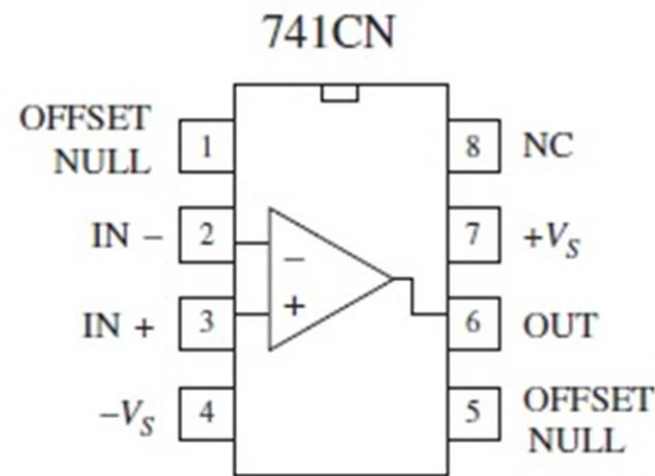
Electrical Systems Element

□ Operational Amplifiers (Op-Amp)

- An **operational amplifier (op-amp)** is an **active** element.
- It is basically a **voltage amplifier** with a **very large open-loop gain** when it is operating in its **linear** region.
- Op-amps are available in the form of **IC chip**. **IC 741** is a typical example of a commonly used op-amp IC chip.
- Typically, it consists many **transistors** plus a number of **resistors** and **capacitors**.
- It has two terminals on the input side, and one terminal on the output side.
- The input terminals are called the **inverting** (−) and **noninverting** (+) terminals.



Circuit Symbol



Electrical Systems Element Laws

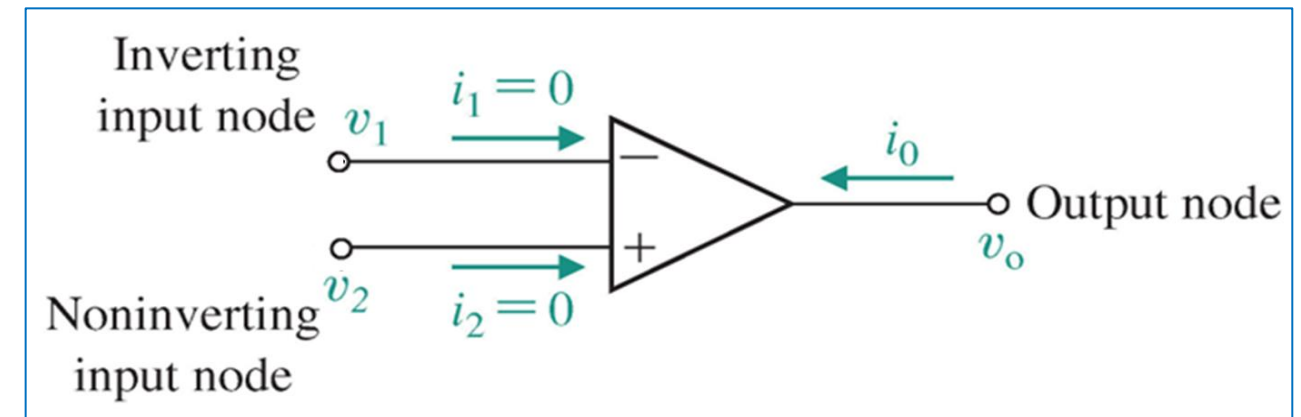
□ Ideal Op-Amps

- Characteristics of an **ideal op-amp** is:

1. High input impedance, ($Z_{in} = 10^5$ to $10^{13} \Omega$, ideally ∞), hence, $i_1 = 0$ and $i_2 = 0$
2. Low output impedance, ($Z_{out} = 10$ to 100Ω , ideally 0),
3. High constant **open-loop gain** amplification, ($A = 10^5$ to 10^8 , ideally ∞)
4. The output voltage is given by,

$$v_o = A(v_2 - v_1)$$

- The output voltage **cannot exceed** the **supply voltage**.



- Op-Amps are the building blocks of **analog** control systems and electronic circuits.
- They are designed to perform some specific circuits when resistors and capacitors are connected to their terminals:
 - To **interface the subsystem** connections,
 - To **implement amplifiers** and **filters**,
 - To perform mathematical operations of **addition, subtraction, multiplication, inversion, differentiation, and integration**.

Modeling of Electrical Systems

□ Inverting Amplifier

- Consider the following **Inverting Amplifier** operating under **ideal** condition.
- Here, R_1 is the **input element** and R_2 is the **feedback element**.
- Since $i_1 = 0$, from a KCL at node v_1 we have:

$$i_{R_1} = i_{R_2} + i_1 \rightarrow \frac{v_{in} - v_1}{R_1} = \frac{v_1 - v_o}{R_2}$$

- We know that,

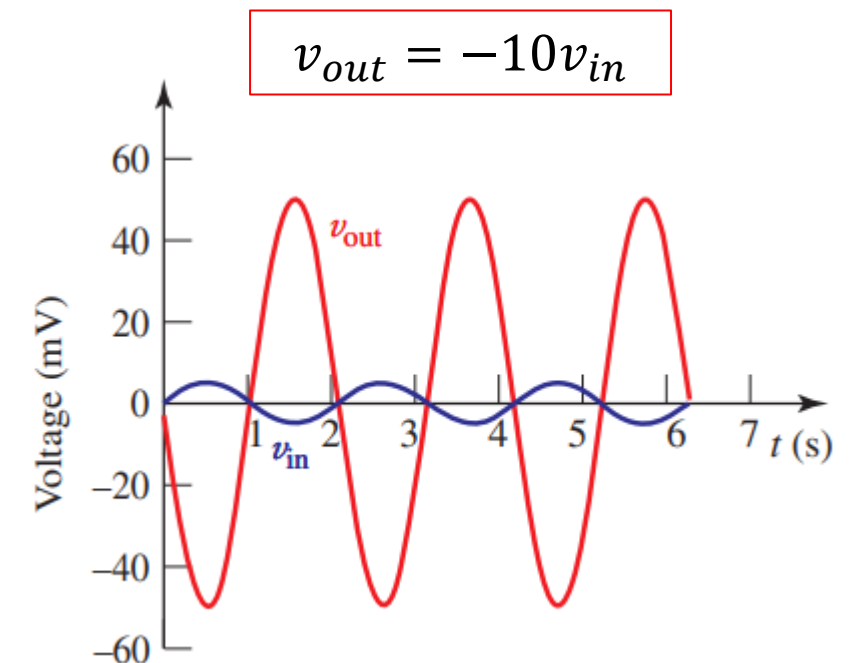
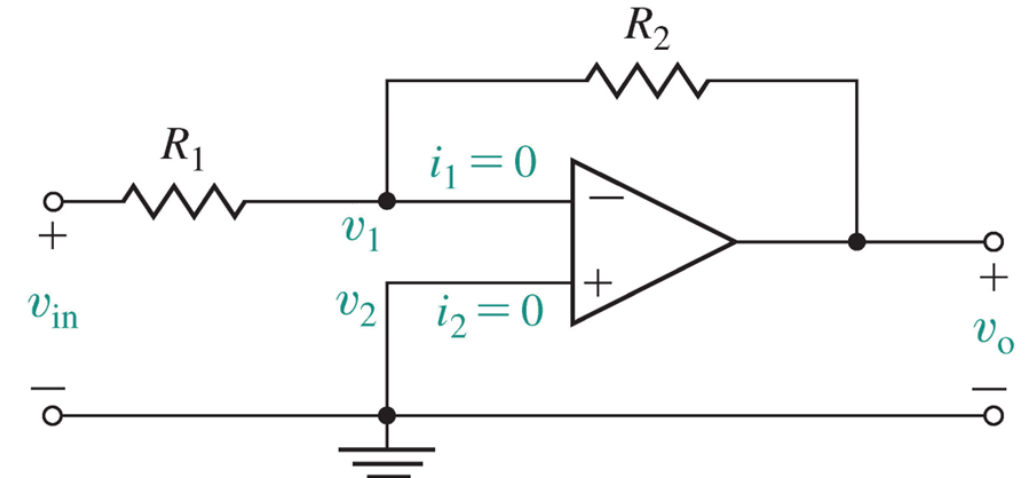
$$v_o = A(v_2 - v_1) \xrightarrow{v_2=0} v_o = -Av_1 \rightarrow v_1 = -\frac{v_o}{A}$$

- Since $A = \infty$, we have $v_1 \approx 0$.

$$\frac{v_{in}}{R_1} = \frac{-v_o}{R_2} \rightarrow v_o = \left(-\frac{R_2}{R_1}\right) v_{in}$$

- Therefore, an inverting amplifier **reverses the polarity** of the input signal.
- The ratio of $-R_2/R_1$ is called the **gain** of the amplifier.
 - If $R_2 > R_1$ this circuit **amplifies** the input voltage signal.
 - If $R_2 < R_1$ the signal will be **attenuated**.
 - If $R_2 = R_1$ the circuit will be an **inverter**.

$$v_o = -v_{in}$$



The graph shows a typical input-output signal for gain of -10.

Modeling of Electrical Systems

Example 4 In the following op-amp circuit, if $v_i = 0.5\text{ V}$, calculate:
a) The output voltage v_o , b) The current in the $10\text{k}\Omega$ and $25\text{k}\Omega$ resistors.

This is an **inverting** op-amp circuit. The general form of the input-output relation is:

$$v_o = \left(-\frac{R_2}{R_1} \right) v_i = \left(-\frac{25\text{k}\Omega}{10\text{k}\Omega} \right) (0.5\text{V}) = (-2.5)(0.5\text{V}) = -1.25\text{ V}$$

The **current** through the $10\text{k}\Omega$ resistor is

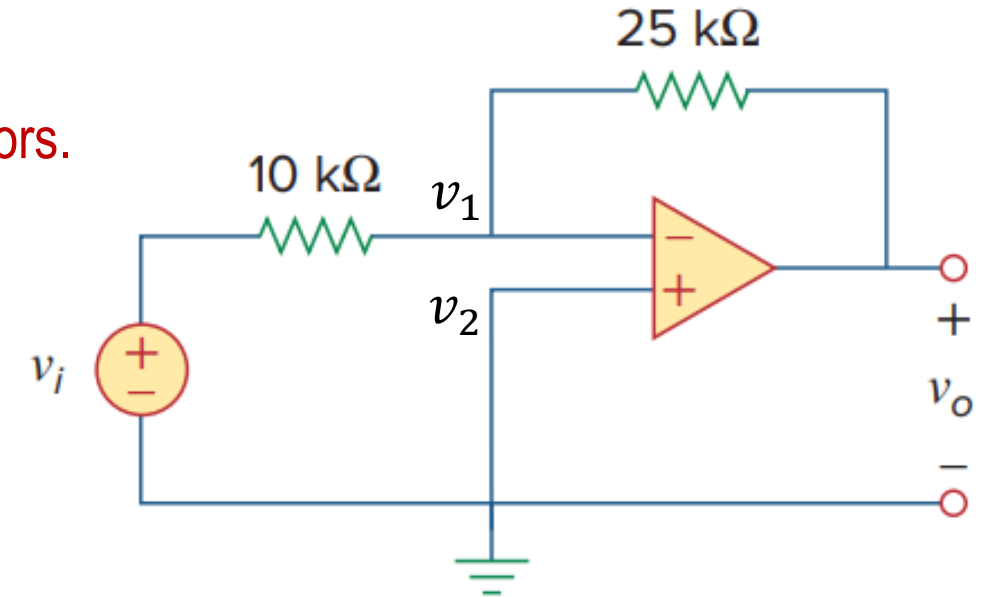
$$i_{R_1} = \frac{v_i - v_1}{10\text{k}\Omega}$$

Since $v_2 = 0$, then $v_1 \approx 0$. Thus,

$$i_{R_1} = \frac{v_i}{10\text{k}\Omega} = \frac{0.5\text{V}}{10\text{k}\Omega} = 0.05\text{ mA} = 50\text{ }\mu\text{A}$$

Apply KCL at node v_1 ,

$$i_{R_1} = i_{R_2} + i_1 \xrightarrow{i_1=0} i_{R_1} = i_{R_2} \rightarrow i_{R_2} = 50\text{ }\mu\text{A}$$



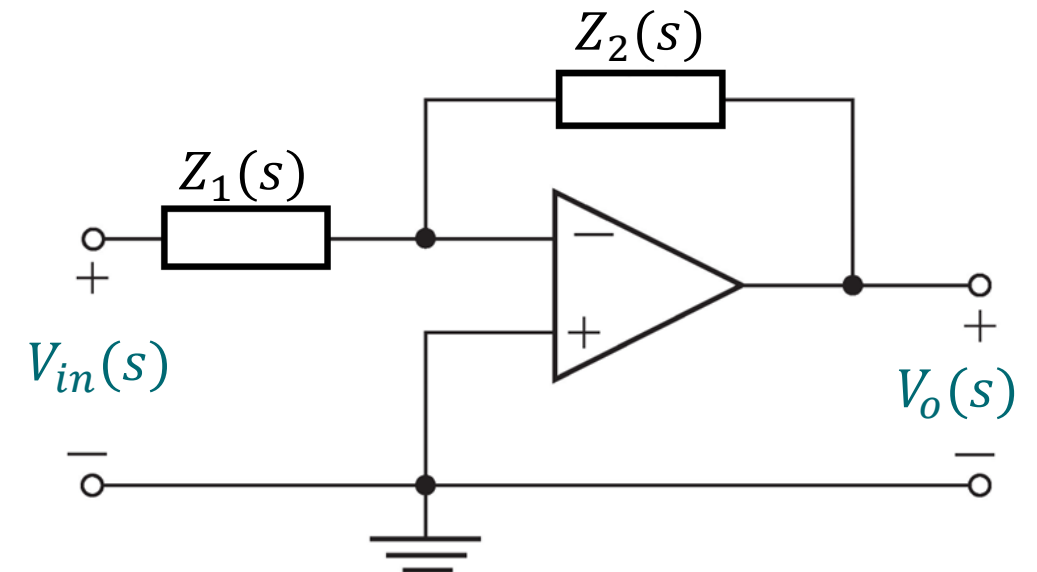
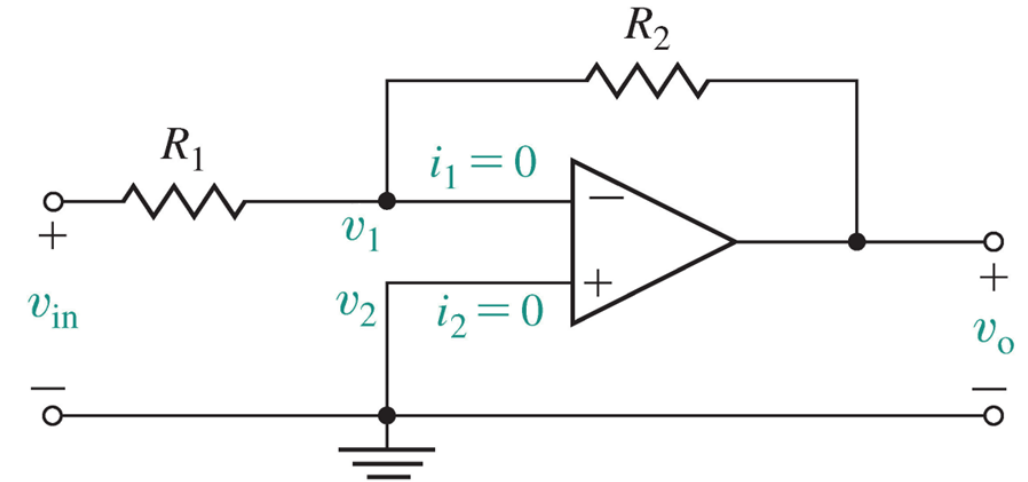
Modeling of Electrical Systems

□ Inverting Amplifier: Transfer Function Model

- We can use the **impedance** concept to obtain a general **transfer function model** for an **inverting amplifier** circuit.
- Assume the inverting amplifier in **Laplace** domain with the general input element of $Z_1(s)$ and feedback element of $Z_2(s)$.
- The **transfer function** model is obtained as:

$$\frac{V_o(s)}{V_{in}(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

Transfer Function Model



Modeling of Electrical Systems

Example 5 Find the transfer function $V_o(s)/V_i(s)$, for the following op-amp circuit.

This is an **inverting** op-amp circuit.

The general form of the transfer function is:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

First, find the **impedances** $Z_1(s)$ and $Z_2(s)$:

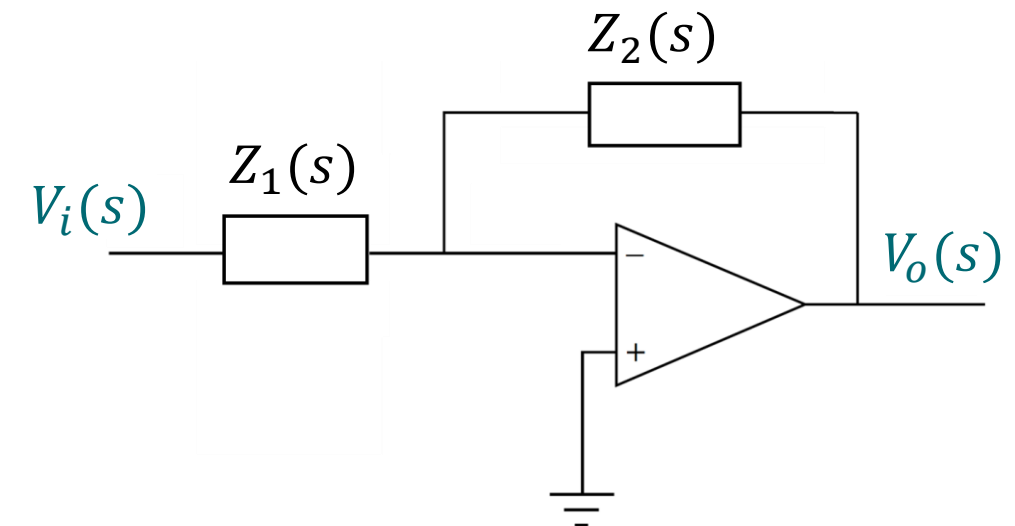
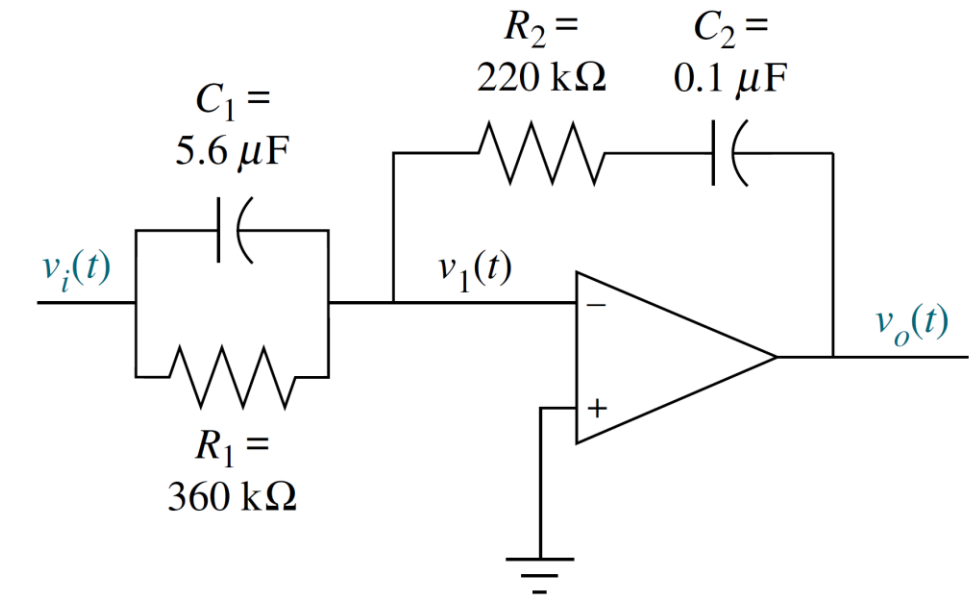
$$Z_1(s) = \frac{R_1 \left(\frac{1}{C_1 s} \right)}{R_1 + \frac{1}{C_1 s}} = \frac{R_1}{R_1 C_1 s + 1}$$

$$Z_2(s) = R_2 + \frac{1}{C_2 s} = \frac{R_2 C_2 s + 1}{C_2 s}$$

The **transfer function** is:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)} = -\frac{\frac{R_2 C_2 s + 1}{C_2 s}}{\frac{R_1}{R_1 C_1 s + 1}} = -\frac{(R_1 C_1 s + 1)(R_2 C_2 s + 1)}{R_1 C_2 s}$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{(2.016s + 1)(0.022s + 1)}{0.036s}$$



Modeling of Electrical Systems

- The general transfer function model of **inverting amplifier** help us to derive the transfer function model of some **mathematical operations**:

$$\frac{V_o(s)}{V_{in}(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

□ Integrating Circuit:

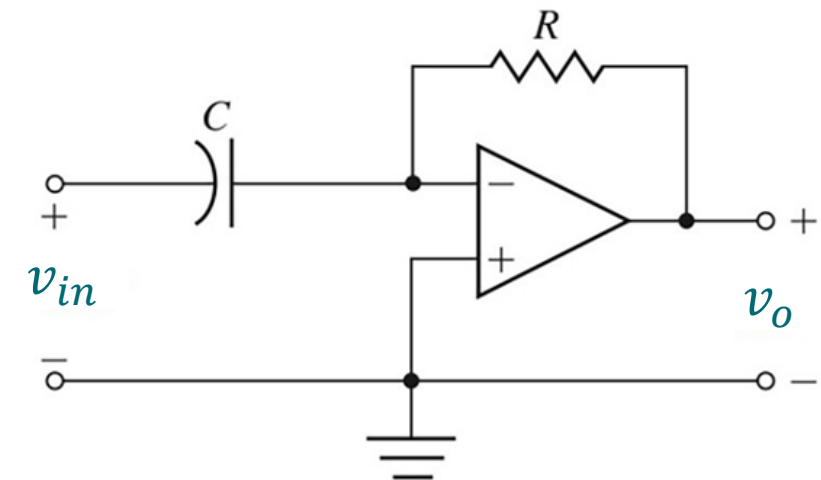
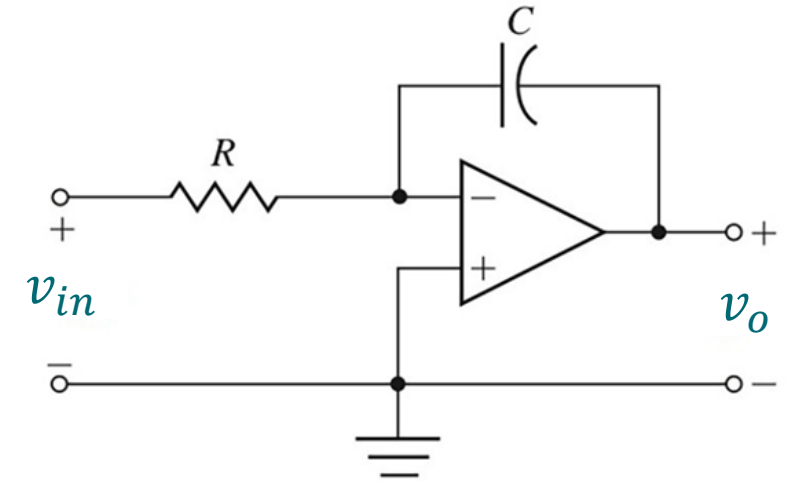
$$\frac{V_o(s)}{V_{in}(s)} = -\frac{1}{Cs} = -\frac{1}{RCs} \rightarrow V_o(s) = -\frac{V_{in}(s)}{RCs} \rightarrow v_o(t) = -\frac{1}{RC} \int_0^t v_{in}(t) dt$$

For example, if $R = 1M\Omega$ and $C = 1\mu F \rightarrow RC = (10^6\Omega)(10^{-6}F) = 1 \text{ sec}$

$$\frac{V_o(s)}{V_{in}(s)} = -\frac{1}{s} \rightarrow v_o(t) = -\int_0^t v_{in}(t) dt$$

□ Differentiating Circuit:

$$\frac{V_o(s)}{V_{in}(s)} = -\frac{R}{1/Cs} = -RCs \rightarrow V_o(s) = -RCsV_{in}(s) \rightarrow v_o(t) = -RC \frac{dv_{in}(t)}{dt}$$



Modeling of Electrical Systems

□ Noninverting Amplifier

- Consider the following **Noninverting Amplifier** operating under the **ideal** condition.

- We have $\rightarrow v_{in} = v_2$

- Since $i_1 = 0$, voltage at node v_1 is $\rightarrow v_1 = \frac{R_1}{R_1 + R_2} v_o$

- We know that,

$$v_o = A(v_2 - v_1) \rightarrow v_o = A\left(v_{in} - \frac{R_1}{R_1 + R_2} v_o\right) \rightarrow v_o = \left(\frac{A(R_1 + R_2)}{AR_1 + R_1 + R_2}\right) v_{in} = \left(\frac{R_1 + R_2}{R_1 + \frac{1}{A}(R_1 + R_2)}\right) v_{in}$$

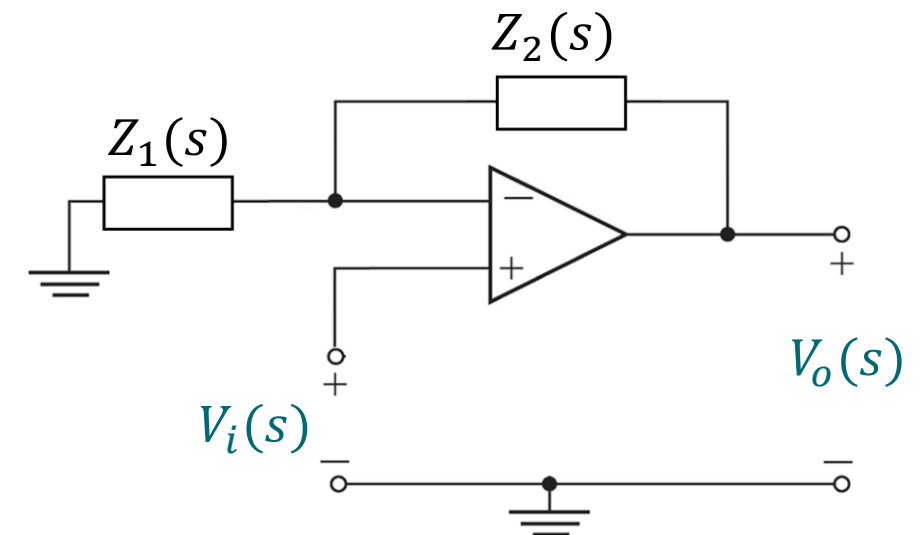
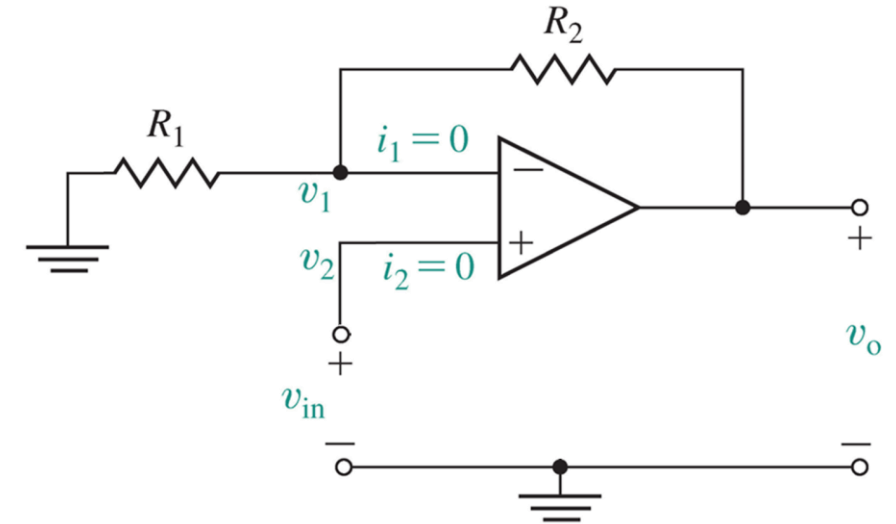
- Since $A = \infty$, we have,

$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_{in}$$

- A **noninverting amplifier** is designed to provide a **positive voltage gain**.
- The general **transfer function** model using the **impedance** concept:

$$\frac{V_o(s)}{V_{in}(s)} = 1 + \frac{Z_2(s)}{Z_1(s)} = \frac{Z_1(s) + Z_2(s)}{Z_1(s)}$$

Transfer Function Model



Modeling of Electrical Systems

Example 6 Find the transfer function $V_o(s)/V_i(s)$, for the following circuit.

This is a **noninverting** op-amp circuit.

The general form of the transfer function is:

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2(s)}{Z_1(s)}$$

First find the **impedances** $Z_1(s)$ and $Z_2(s)$:

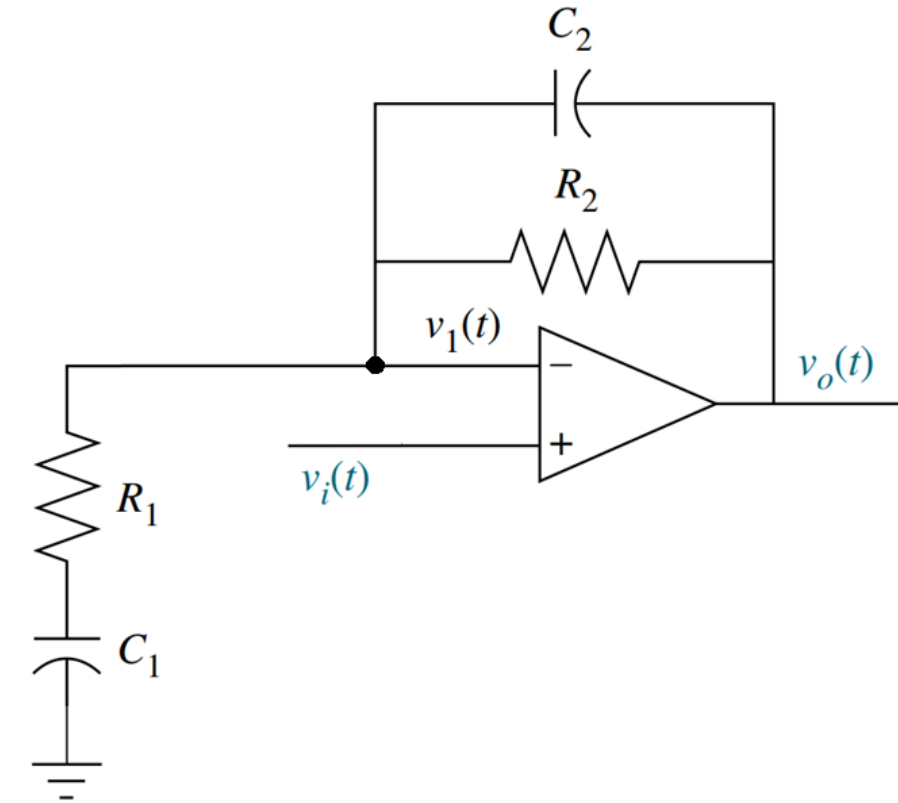
$$Z_1(s) = R_1 + \frac{1}{C_1 s} = \frac{R_1 C_1 s + 1}{C_1 s}$$

$$Z_2(s) = \frac{R_2 \left(\frac{1}{C_2 s} \right)}{R_2 + \frac{1}{C_2 s}} = \frac{R_2}{R_2 C_2 s + 1}$$

The **transfer function** is:

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2(s)}{Z_1(s)} = 1 + \frac{\frac{R_2}{R_2 C_2 s + 1}}{\frac{R_1 C_1 s + 1}{C_1 s}} = 1 + \frac{R_2 C_1 s}{(R_1 C_1 s + 1)(R_2 C_2 s + 1)} = \frac{(R_1 C_1 s + 1)(R_2 C_2 s + 1) + R_2 C_1 s}{(R_1 C_1 s + 1)(R_2 C_2 s + 1)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}{R_1 R_2 C_1 C_2 s^2 + (R_1 C_1 + R_2 C_2) s + 1}$$



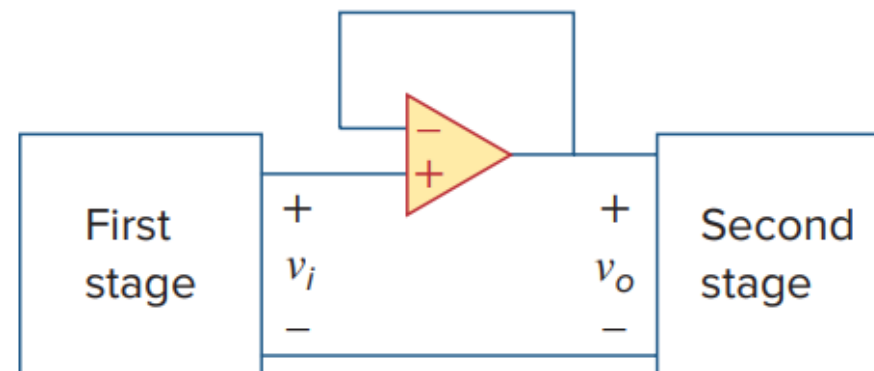
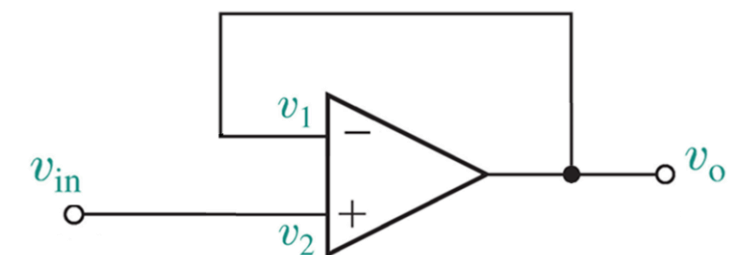
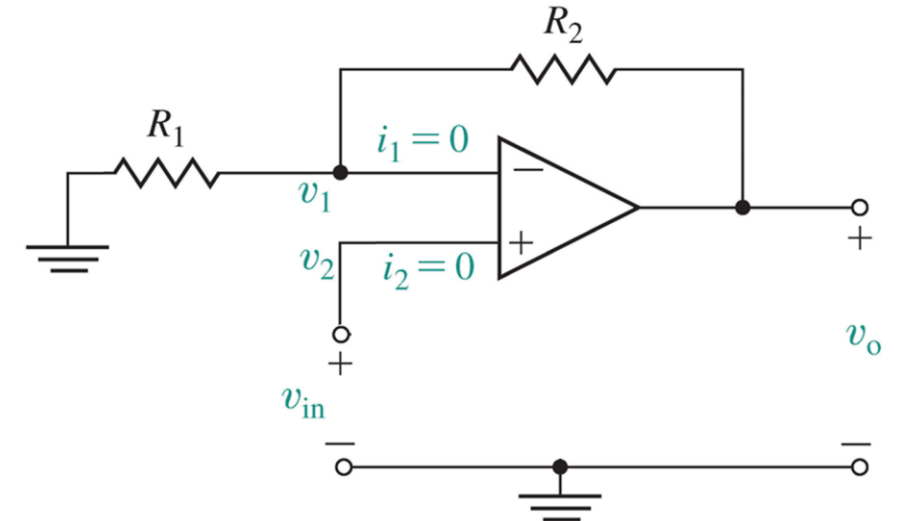
Modeling of Electrical Systems

□ Voltage Follower

- A **Voltage Follower** is implemented by setting $R_2 = 0$ (**short circuit**) and $R_1 = \infty$ (**open circuit**) in the **noninverting amplifier** circuit.
- The voltage follower has a direct **negative feedback**, which means the output is directly connected to its inverting input.
- The input-output relationship of a **voltage follower** operating under the ideal condition is:

$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_{in} \xrightarrow[R_2 = 0]{R_1 = \infty} \boxed{v_o = v_{in}}$$

- The voltage follower has a **very high input impedance** and is therefore useful as a **buffer** to **isolate** one circuit from another.
- The voltage follower minimizes interaction between the two stages and eliminates interstage **loading effect**.



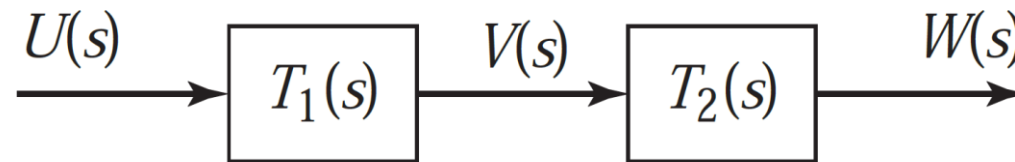
Modeling of Electrical Systems

□ Loading Effect & Block Diagram Models

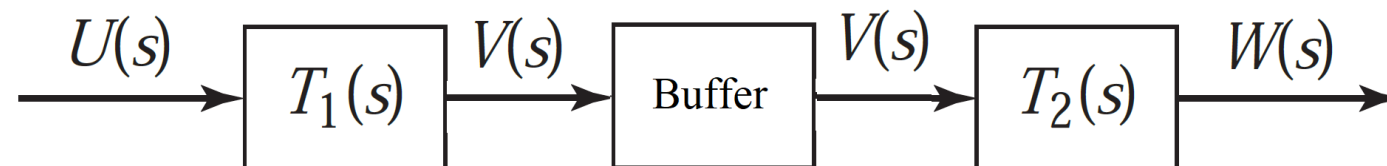
- Suppose two **passive elements/circuits** whose individual transfer functions are $T_1(s)$ and $T_2(s)$ are physically connected **end-to-end** so that the output of the **lefthand** element becomes the input to the **righthand** element.

$$T_1(s) = \frac{V(s)}{U(s)}$$

$$T_2(s) = \frac{W(s)}{V(s)}$$



- We can represent this connection by the **series block diagram** shown as below **only if** the output w of the **righthand** element does **not affect** the inputs u and v or the behavior of the **lefthand** element.
- If it does, the **righthand** element is said to **"load"** the **lefthand** element.
- We can avoid the loading effect by connecting the two stages via a **voltage buffer** to **isolate** the stages.

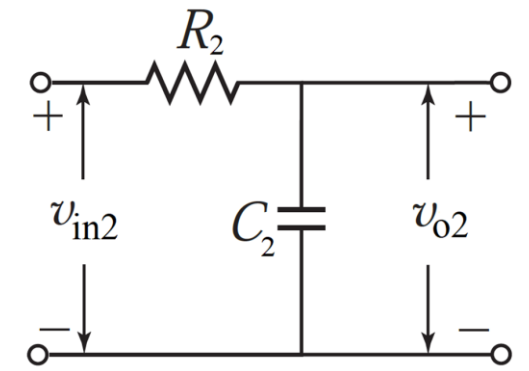
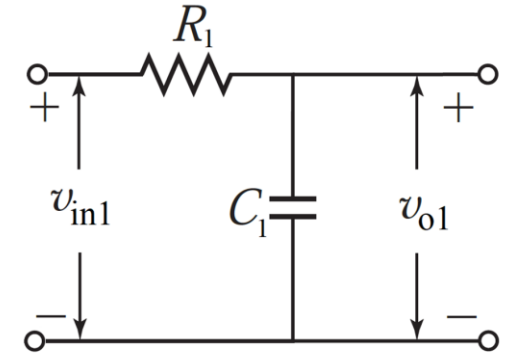


Modeling of Electrical Systems

Example 7 Consider the following two electric circuits with their individual transfer functions $G_1(s)$ and $G_2(s)$.

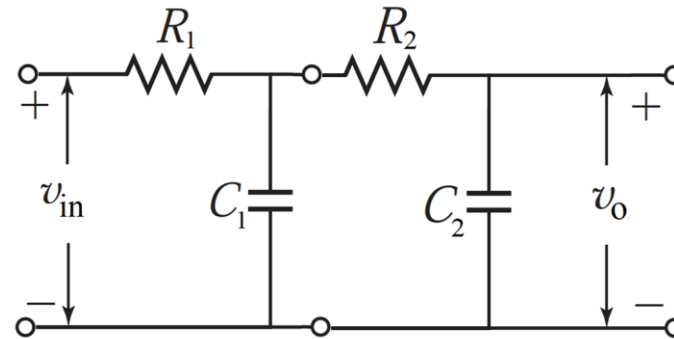
$$G_1(s) = \frac{V_{o1}(s)}{V_{in1}(s)} = \frac{1}{R_1 C_1 s + 1}$$

$$G_2(s) = \frac{V_{o2}(s)}{V_{in2}(s)} = \frac{1}{R_2 C_2 s + 1}$$

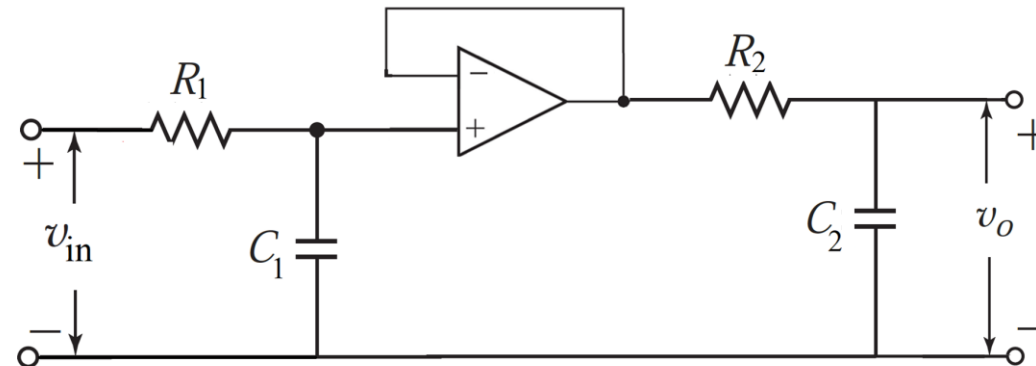


Find the overall transfer function $V_o(s)/V_{in}(s)$ in each cases and determine which implementation can be shown as a series connected block diagram.

(a) The circuits are connected end-to-end as shown



(b) The circuits are connected via a voltage buffer stage as shown:



Modeling of Electrical Systems

Example 7

(a) The circuits are connected end-to-end as shown below:

First, convert the circuit to Laplace domain based on the impedances of the elements:

Apply a KCL at node $V_1 \rightarrow I_{R_1}(s) = I_{C_1}(s) + I_{R_2}(s)$

$$\frac{V_{in}(s) - V_1(s)}{R_1} = \frac{V_1(s)}{\frac{1}{C_1 s}} + \frac{V_1(s) - V_o(s)}{R_2}$$

$$R_2(V_{in}(s) - V_1(s)) = R_1 R_2 C_1 s V_1(s) + R_1(V_1(s) - V_o(s))$$

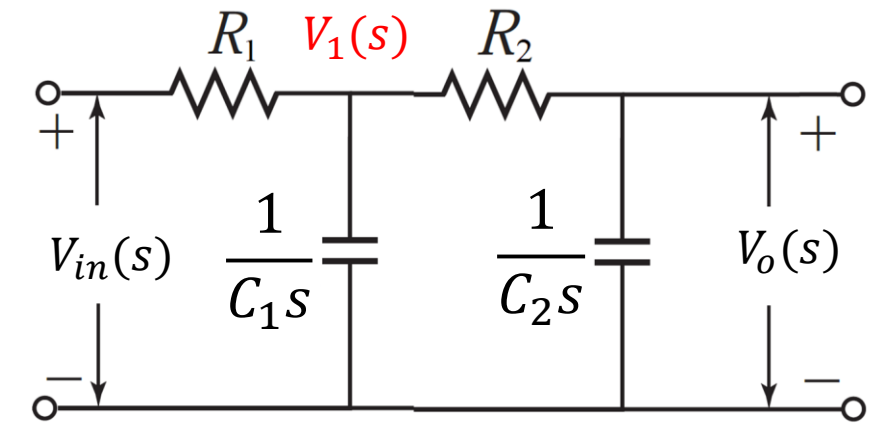
$$R_2 V_{in}(s) + (-R_2 - R_1 R_2 C_1 s - R_1) V_1(s) + R_1 V_o(s) = 0$$

$$R_2 V_{in}(s) + (-R_2 - R_1 R_2 C_1 s - R_1)(R_2 C_2 s + 1) V_o(s) + R_1 V_o(s) = 0$$

$$R_2 V_{in}(s) + (-R_2^2 C_2 s - R_2 - R_1 R_2^2 C_1 C_2 s^2 - R_1 R_2 C_2 s - R_1 R_2 C_2 s - R_1 + R_1) V_o(s) = 0$$

$$V_{in}(s) + (-R_2 C_2 s - 1 - R_1 R_2 C_1 C_2 s^2 - 2R_1 C_2 s) V_o(s) = 0$$

$$\frac{V_o(s)}{V_{in}(s)} = \frac{1}{R_1 R_2 C_1 C_2 s^2 + (R_2 C_2 + 2R_1 C_2)s + 1}$$



$$G_1(s) = \frac{1}{R_1 C_1 s + 1} \quad G_2(s) = \frac{1}{R_2 C_2 s + 1}$$

Apply voltage-division:

$$V_o(s) = \frac{\frac{1}{C_2 s}}{R_2 + \frac{1}{C_2 s}} V_1(s)$$

$$V_1(s) = (R_2 C_2 s + 1) V_o(s)$$

The overall transfer function is **not identical** with $G_1(s)G_2(s)$ because of the loading effect.

We cannot show the overall system as the series block diagram.

Modeling of Electrical Systems

Example 7

(b) The circuits are connected via an op-amp buffer stage as shown below:

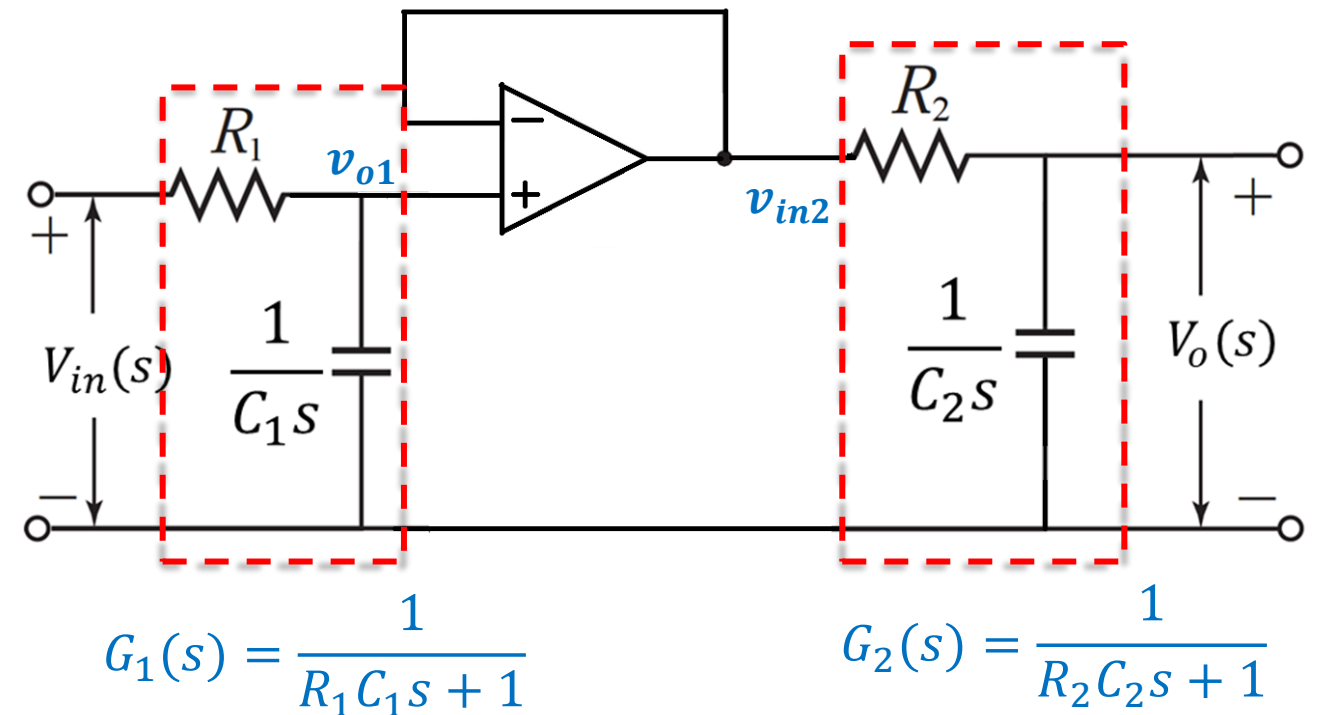
Since the circuit stages are connected via an **op-amp buffer** we can show there is no loading effect between the stages.

$$v_{in2} = v_{o1}$$

The overall system can be modeled by the following block diagram model:



$$\frac{V_o(s)}{V_{in}(s)} = \frac{V_o(s)}{V_{in2}(s)} \cdot \frac{V_{in2}(s)}{V_{o1}(s)} \cdot \frac{V_{o1}(s)}{V_{in}(s)} = G_2(s) \cdot 1 \cdot G_1(s)$$



$$\frac{V_o(s)}{V_{in}(s)} = \frac{1}{(R_1 C_1 s + 1)(R_2 C_2 s + 1)}$$

THANK YOU